

LOAN APPROVAL INTELLIGENCE

An AI-Powered Data Analytics and Prediction System for Smarter Loan Decisions

AICTE | IBM SkillsBuild Data Analytics with AI Internship Program 2026

Project Report

This report documents the complete analysis of historical loan applications, from data preparation and exploratory analysis through machine learning, model evaluation, feature importance, and business interpretation.

1. Executive Summary

Loan Approval Intelligence analyzes 4,269 historical loan applications to identify patterns associated with approval and rejection and to evaluate machine-learning approaches for predicting loan status. The dataset contains applicant, loan, credit-score, and asset information.

There were 2,656 approved applications (62.22%) and 1,613 rejected applications (37.78%). CIBIL score showed the clearest observed separation between the two groups. In the fitted Random Forest model, CIBIL score represented approximately 80.55% of total feature importance.

Random Forest achieved 99.53% accuracy and 99.96% ROC-AUC on the held-out test set. Five-fold cross-validation produced mean accuracy of 99.60% with a standard deviation of 0.12 percentage points. These results are specific to this dataset and do not establish causality or guarantee performance on external lending populations.

2. Business Problem and Objectives

The project examines whether historical loan-application data can provide useful decision-support insights for understanding loan approval outcomes.

- Measure approval and rejection KPIs.
- Identify patterns in credit, loan, applicant, and asset variables.
- Create useful derived financial features.
- Compare Logistic Regression, Decision Tree, and Random Forest.
- Evaluate predictive performance using multiple metrics.
- Identify important predictive features and translate them into business insights.

3. Dataset and Preparation

The dataset contains 4,269 rows and 13 original columns. The target is loan_status with Approved and Rejected outcomes. The original variables cover dependents, education, self-employment, income, loan amount, loan term, CIBIL score, four asset categories, and loan status.

- Column names and categorical values were stripped of unnecessary whitespace.
- Missing values: 0.
- Duplicate rows: 0.
- loan_id was excluded from model features because it is an identifier.
- Categorical variables were encoded using One-Hot Encoding inside a preprocessing pipeline.

4. Feature Engineering

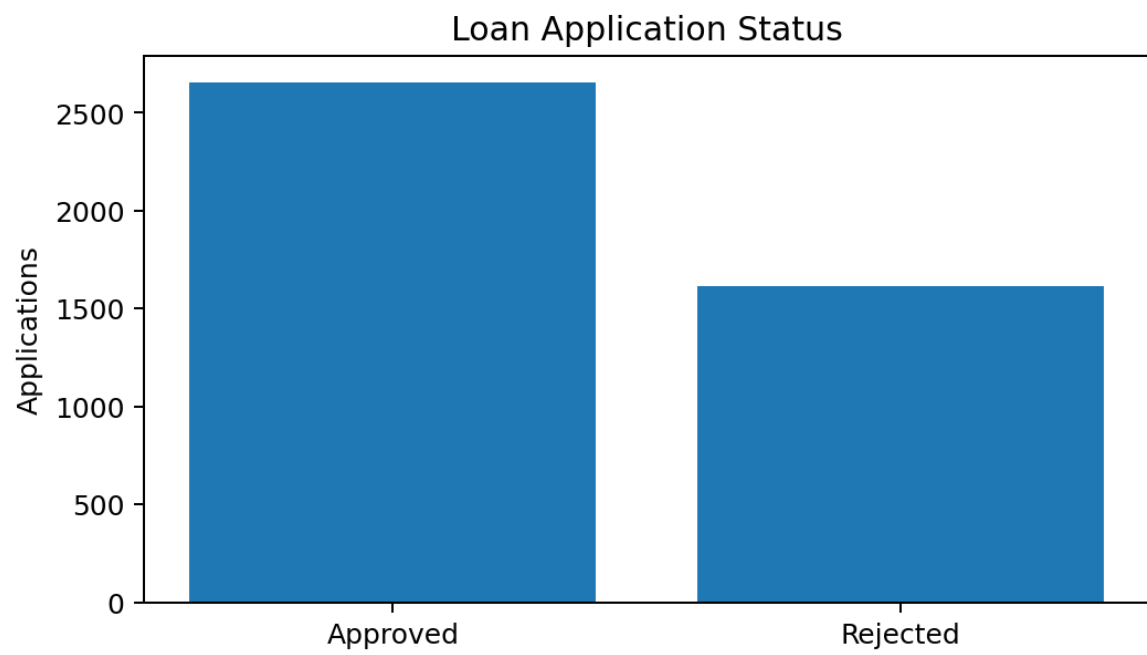
Total Assets = residential + commercial + luxury + bank assets.

Loan-to-Income Ratio = loan amount / annual income.

5. Key KPIs

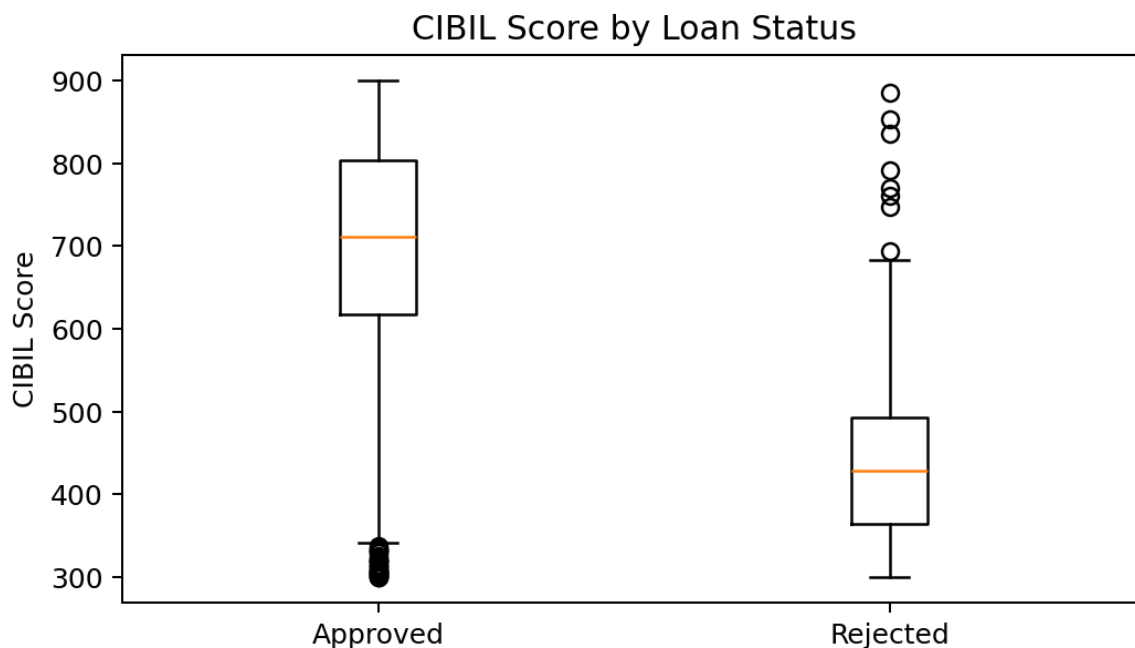
Metric	Value
Total applications	4,269
Approved	2,656
Rejected	1,613
Approval rate	62.22%
Rejection rate	37.78%
Average income	■50,59,123.92

Average loan	■1,51,33,450.46
Average CIBIL	599.94



6. Exploratory Data Analysis

CIBIL Score



The median CIBIL score was 711 for approved applications and 429 for rejected applications, showing the strongest visible separation in the EDA.

Other Findings

- Annual income was very similar across approval groups: median ₹50 lakh for approved versus ₹51 lakh for rejected.
- Median loan amount was also similar: ₹1.46 crore for approved versus ₹1.45 crore for rejected.
- Median loan term was 10 years for approved and 12 years for rejected, showing some difference.
- Median total assets were similar: ₹3.10 crore for approved versus ₹3.19 crore for rejected.
- Education, self-employment, and number of dependents showed relatively small differences in approval rates.

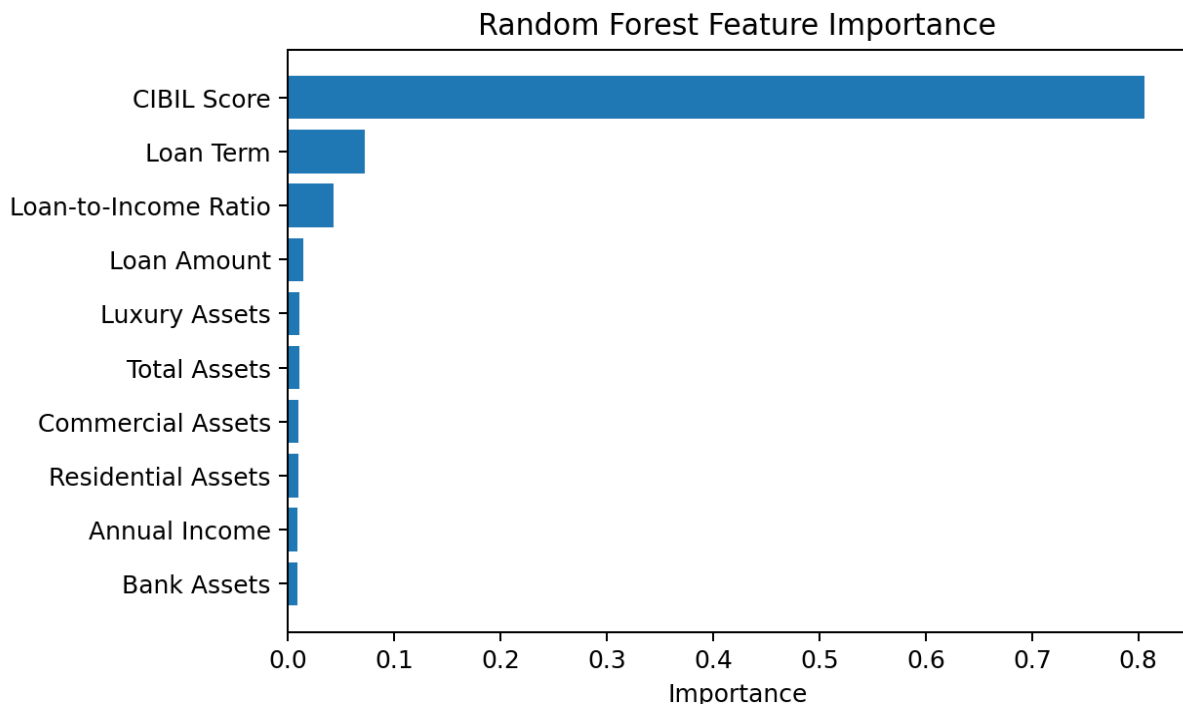
7. Machine Learning Methodology

The target was encoded as Approved = 1 and Rejected = 0. The data was split into 80% training (3,415 records) and 20% testing (854 records), using stratification. Three models were trained: Logistic Regression, a depth-limited Decision Tree, and Random Forest with 200 estimators and maximum depth 10.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	80.68%	79.80%	92.28%	85.59%	88.63%
Decision Tree	99.41%	99.25%	99.81%	99.53%	99.43%
Random Forest	99.53%	99.25%	100.00%	99.62%	99.96%

Random Forest produced the strongest test-set metrics among the evaluated models. Five-fold cross-validation gave accuracy scores of 99.41%, 99.53%, 99.77%, 99.65%, and 99.65%, with mean 99.60% and standard deviation 0.12 percentage points.

8. Feature Importance and Explainability



CIBIL score had approximately 80.55% feature importance in the fitted Random Forest, followed by loan term at 7.19% and loan-to-income ratio at 4.28%. The remaining variables had substantially smaller individual importance values.

Feature importance describes the contribution of variables to the fitted model's predictive behavior; it does not establish that a variable causes approval or rejection.

9. Business Insights and Recommendations

- **Credit-related information is central in this dataset.** CIBIL score was both strongly separated in EDA and dominant in the Random Forest feature-importance analysis.
- **Loan structure provides additional information.** Loan term and loan-to-income ratio contributed secondary predictive information.
- **Several variables were weak standalone separators.** Income, loan amount, assets, education, self-employment, and dependents showed considerable overlap between the two outcome groups.
- **Use model outputs as decision support.** Predictions should complement, rather than automatically replace, institutional lending review and relevant controls.
- **Validate externally before deployment.** Testing on external and more recent datasets is necessary before drawing conclusions about real-world performance.

10. Limitations

- The analysis uses a single historical dataset.
- Observed associations are not causal conclusions.
- Very high tree-model performance may reflect strong structure specific to this dataset.
- External validation is required before generalization.
- Broader operational, regulatory, and policy variables are outside the dataset.

11. Conclusion

The project demonstrates an end-to-end workflow from raw loan data to business insights and predictive modeling. CIBIL score emerged as the dominant predictive feature, while loan term and loan-to-income ratio provided additional predictive

information. Random Forest achieved 99.53% test accuracy and 99.60% mean five-fold cross-validation accuracy on this dataset.

The findings are dataset-specific. Any real-world lending application would require external validation, governance, fairness assessment, monitoring, and appropriate human oversight.

12. Submission Deliverables

File	Purpose
loan_approval_intelligence.py	Complete analytics and ML pipeline
requirements.txt	Python dependencies
README.md	Project documentation and run instructions
Project_Report.pdf	Formal project report